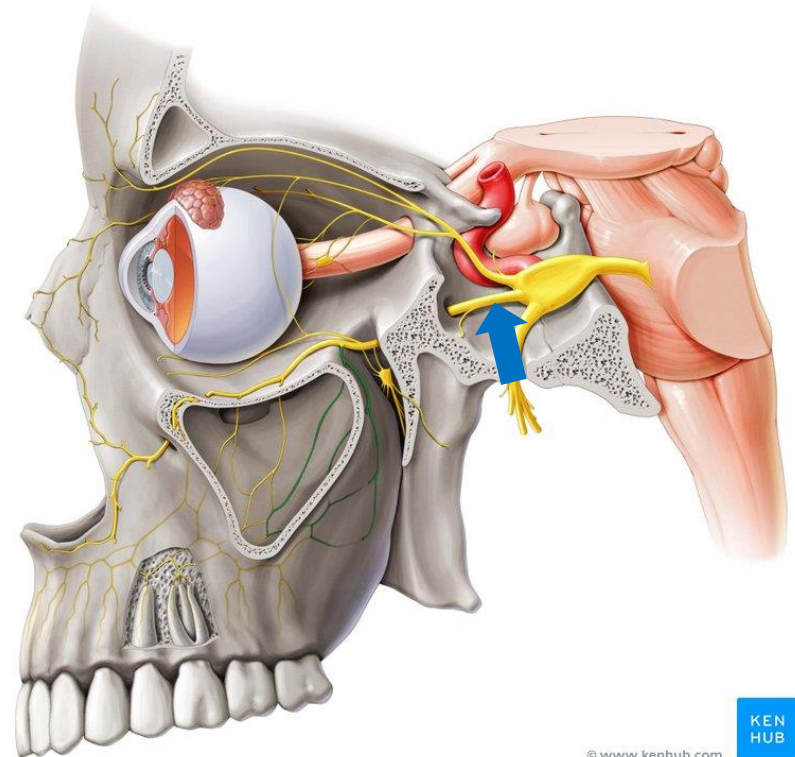


Maxillary nerve (V2):

- It is sensory in nature.
- It arises from the trigeminal ganglion.
- Then runs in the lateral wall of cavernous sinus.
- Exits the skull through foramen rotundum to reach the pterygopalatine fossa.
- In the pterygopalatine fossa it runs in the upper part & continues as infraorbital nerve.
- In the fossa, it is closely related to the pterygopalatine ganglion and gives off the zygomatic & posterior superior alveolar branches.



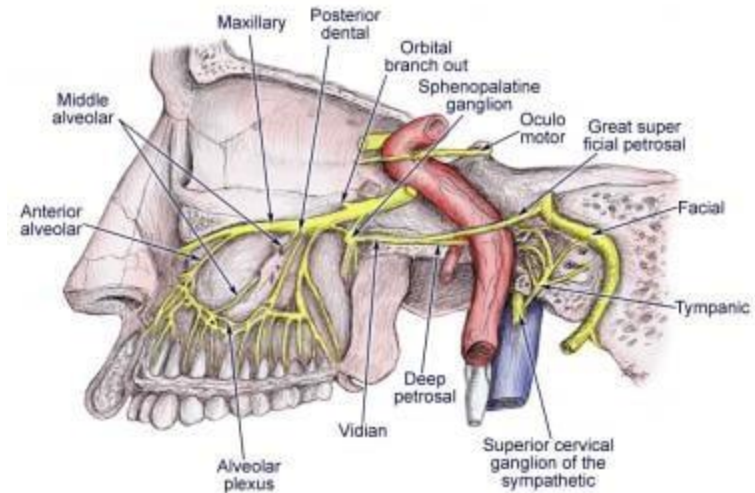
Branches of the maxillary nerve:

1. Posterior superior alveolar nerve:

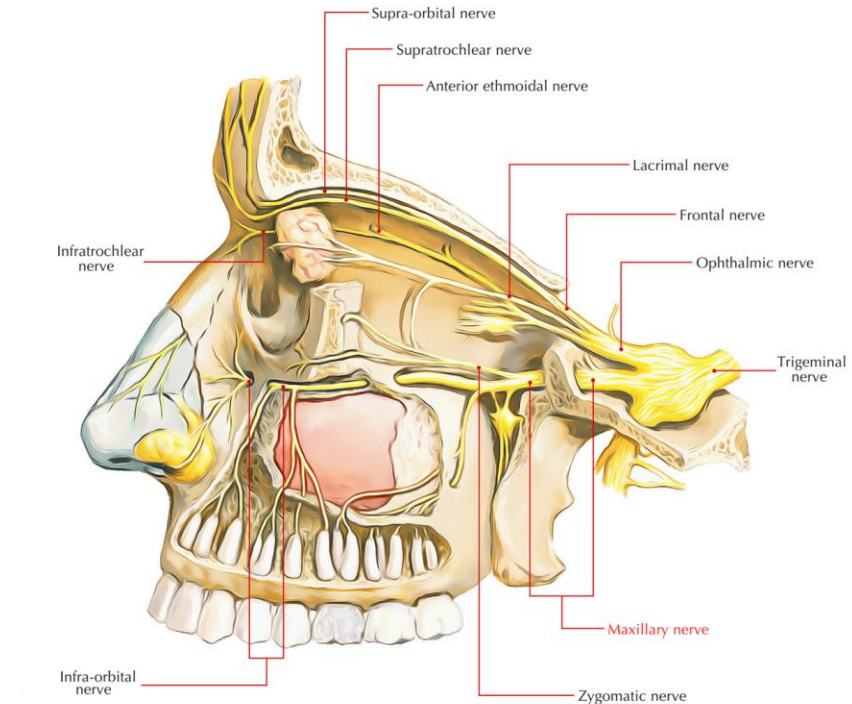
- Enters the body of maxilla posteriorly and supplies the upper three molar teeth & adjacent part of gum.

2. Zygomatic nerve:

- This branch also arises in the pterygopalatine fossa.
- Enters the orbit through inferior orbital fissure.
- In the orbit, it runs along the lateral wall and enters the zygomatic bone through a foramen.
- In the substance of zygomatic bone, it divides into the zygomaticotemporal & zygomaticofacial branches.

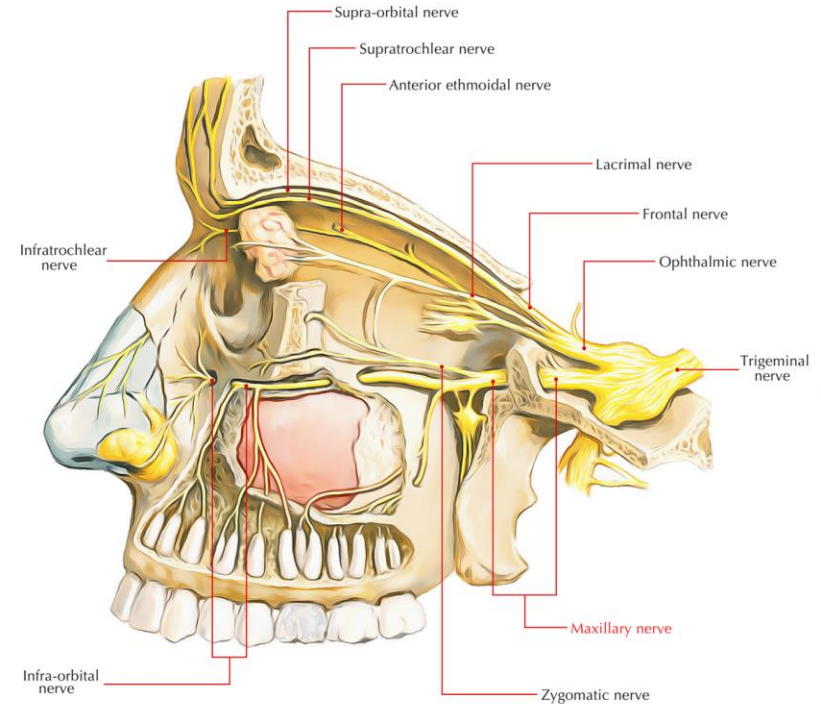


- The zygomaticotemporal & zygomaticofacial nerves supply the skin of face and anterior part of temple.
- A communicating branch to the lacrimal nerve may arise from the zygomaticotemporal nerve.



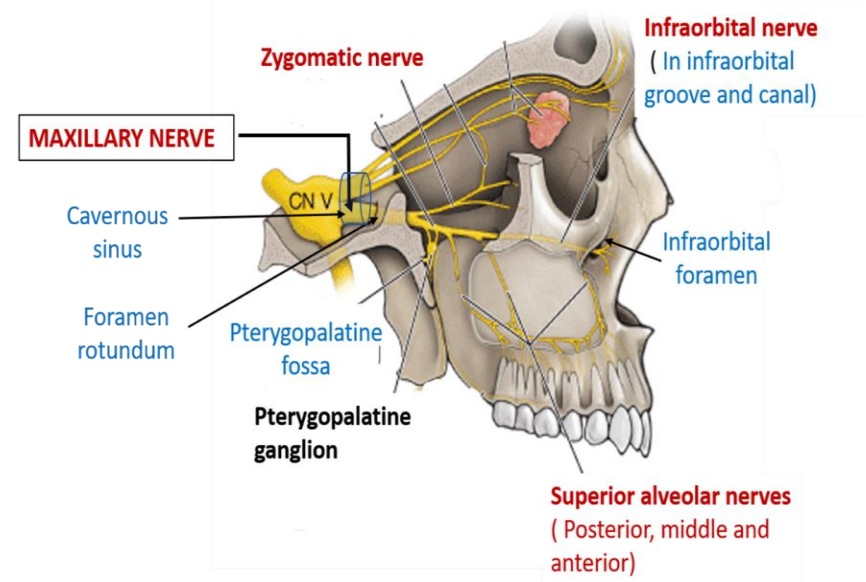
3. Infraorbital nerve:

- It is a continuation of maxillary nerve.
- Enters the orbit through inferior orbital fissure.
- At first runs on the floor of orbit and then enters the infraorbital canal.
- It emerges on face via infraorbital foramen.
- Terminates by splitting into palpebral, nasal, & labial branches.



Branches of infraorbital nerve:

- i. **Middle superior alveolar nerve:** supplies the upper premolar teeth.
- ii. **Anterior superior alveolar nerve:** supplies the upper incisor & canine teeth, and lower part of nasal cavity.
- iii. **Terminal branches (palpebral, nasal, & labial):** supply a big part of skin of face and mucous membrane of upper lip & cheek.



Distribution of maxillary nerve:

REFERENCE HANDOUT: TRIGEMINAL NERVE BRANCHES (NOT INCLUDING HITCHHIKING PATHWAYS OF VII, IX) zili@musom.2013

V1 Ophthalmic - GSA (Sensory) only - through Superior Orbital Fissure to Orbit

Nerve	Branches	Innervates
1. Frontal Nerve	a. Supraorbital Nerve b. Supratrochlear Nerve	Scalp forehead, upper eyelid Scalp forehead, upper eyelid Upper eyelid
2. Lacrimal Nerve		Cornea of eye
3. Nasociliary Nerve	a. Long Ciliary Nerve b. Ant. and Post. Ethmoidal Nerves c. Infratrochlear Nerve	Nasal cavity, ethmoid sinus, tip of nose Upper eyelid, nose

V2 Maxillary - GSA (Sensory) only - through Foramen Rotundum to Pterygopalatine Fossa

Nerve	Branches	Innervates
1. Meningeal branches		Dura of mid. Cranial fossa
2. Ganglionic branches	a. Greater Palatine Nerve b. Lesser Palatine Nerve c. Nasopalatine Nerve d. Nasal branches	Hard Palate Soft Palate Nasal Cavity, Hard Palate Nasal Cavity
3. Post. Sup. Alveolar Nerve		Maxillary teeth
4. Infraorbital nerve	a. Ant. Sup. Alveolar Nerve b. Mid. Sup. Alveolar Nerve	Lower eyelid, nose, upper lip Maxillary teeth
5. Zygomatic nerve	a. Zygomaticofacial Nerve b. Zygomaticotemporal Nerve	Skin of cheek Skin of temporal region

V3 Mandibular - GSA (Sensory) and SVE (Motor) - Foramen Ovale to Infratemporal Fossa

Nerve	Branches	Innervates
1. Nervous spinosus		Sensory to Dura of mid Cranial fossa
2. Motor branches		Motor to Med. Pterygoid, Tens. Tympani, Tensor Palati
3. Anterior division	a. Nerve to Lateral Pterygoid b. Masseteric Nerve c. Deep Temporal Nerve d. Buccal Nerve	Motor to Lateral Pterygoid Motor to Masseter Motor to Temporalis Sensory to Cheek
4. Posterior Division	a. Auriculotemporal Nerve b. Lingual Nerve c. Inferior Alveolar Nerve i. Nerve to Mylohyoid ii. Mental Nerve	Sensory to external auditory meatus, tympanic membrane, TMJ, lateral scalp Sensory (touch) ant. 2/3 tongue Sensory to Mandibular teeth Motor to Mylohyoid, ant. Digastric Sensory to Chin, Lower lip

